RETHINKING AUTISM DETECTION: A NEUROPHYSIOLOGICAL APPROACH USING EARLY EEG SIGNALS

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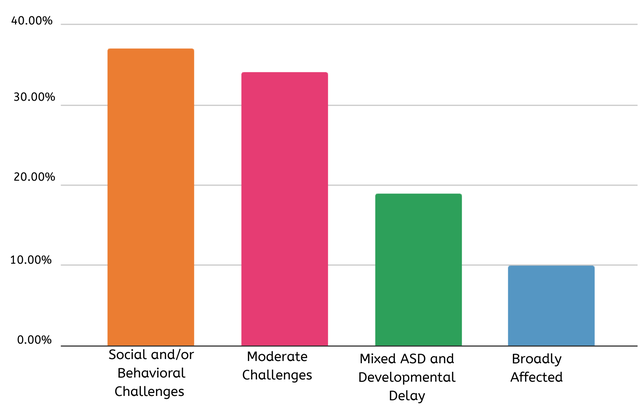
Introduction

Autism Spectrum Disorder (ASD) is currently diagnosed through behavioral observation, often missing the earliest windows for intervention. An electroencephalogram (EEG) is a safe, non-invasive, and low-cost method for earlier ASD detection.[2] This study proposes using EEG to identify neural patterns associated with ASD and to predict the Calibrated Severity Score (CSS, scored 1-10). Further, we propose applying the nonlinear dynamical features extracted from EEG time-series data to Topic Supervised Non-Negative Matrix Factorization (TS-NMF) to predict the four ASD subtypes.[3] Finally, we aim to investigate William Bosl's pipeline for detecting atypical neural dynamics that precede behavioral symptoms and are detectable in the early stages of life through longitudinal EEG evaluation.

Distribution of Four Autism Subtypes



Objective

This is a hypothetical research framework. The goals are to determine whether nonlinear EEG features in early infancy (between three and thirty-six months) can signal atypical brain dynamics before the appearance of behavioral symptoms, distinguish between the four ASD subtypes, and quantify severity using CSS. The broader aim is to establish EEG as a scalable tool for earlier ASD identification and more targeted interventions.

Data Collection

Building off previously collected EEG recordings from William Bosl's study with Boston Children's Hospital, the recordings were gathered longitudinally from infants aged three to thirty-six months.[2] A dynamical systems approach is used to model how brain activity evolves across early development. Participants are initially classified into binary groups - ASD vs. non-ASD, and symptom severity is standardized using CSS.

Methodology

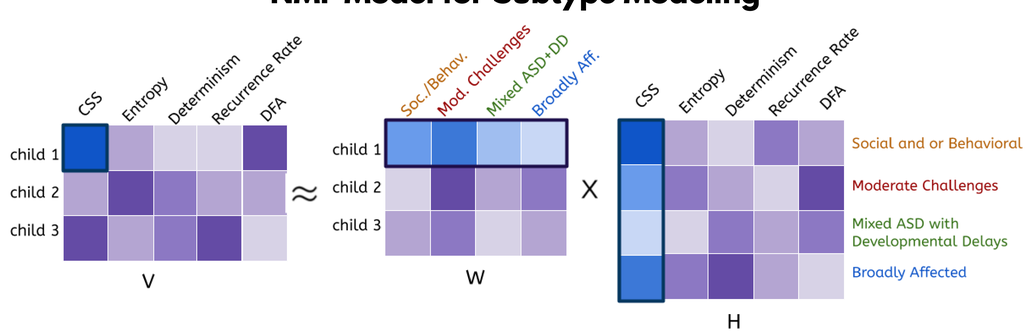
1. EEG recordings are gathered longitudinally from infants aged 3 to 36 months, building on William Bosl's study with Boston Children's Hospital. Participants include high-risk infants (with autistic siblings) and low-risk controls.[2]

2. EEG time-series data is decomposed into frequency bands, from which nonlinear dynamical features are extracted to capture brain complexity and temporal structure.[1] Recursive feature elimination selects the most informative features, which are then fed into a Support Vector Machine (SVM). Output includes binary ASD/non-ASD and CSS reflecting symptom severity.

3. Children on the autism spectrum vary in their neural and behavioral profiles. Clustering before making subtype prediction allows the model to account for individual variability, allowing for more realistic and personalized diagnoses.

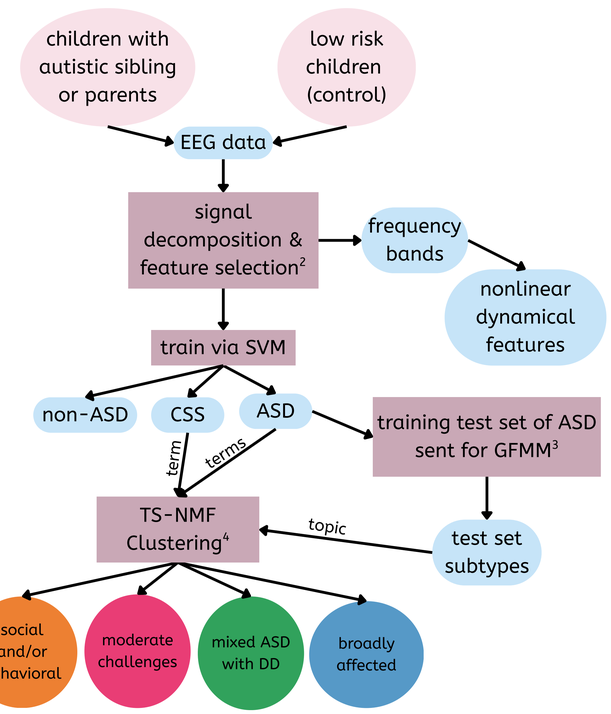
4. SVM-identified ASD participants are clustered into one of four empirically derived subtypes using TS-NMF.[4] Dynamical EEG features from the input matrix, subtype labels from a Generative Finite Mixture Model (GFMM), and initialize the topic labels. Each participant's subtype is assigned by the dominant coefficient in the NMF output, producing an interpretable, data-driven subtype profile for earlier intervention planning.

NMF Model for Subtype Modeling



Note: features listed as "terms" are a small example set that does not reflect the full model

Framework of Diagnosis Process



Conclusion

This framework proposes a path toward personalized, bias-reduced ASD diagnosis. By integrating EEG-based biomarkers with TS-NMF subtype indentification, earlir and more trargtedt interventions become possible, ideally before standard diagnosis around age three. With clinical validation, this approach could scale to universal screening at 12 to 18 months.

Acknowledgements

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